Complications of Liver Disease (Cirrhosis)

Ascites • Spontaneous bacterial peritonitis • Varices • Hepatic encephalopathy • Hepatorenal syndrome — all key points for the exam + drug summary table

1. Background: Cirrhosis and Portal Hypertension

- **Cirrhosis** = advanced liver fibrosis → **portal hypertension** + **liver failure** (↓ synthesis: albumin, clotting factors).
- **Portal hypertension:** hepatic venous pressure gradient (**HVPG**) > 5 mmHg • **≥ 10 mmHg** = clinically significant → varices, ascites • **≥ 12 mmHg** → **variceal bleeding** risk.
- **Splanchnic vasodilation** (nitric oxide) → ↓ effective arterial volume → activation of **RAAS**, sympathetic system, ADH → **sodium and water retention** (ascites, hyponatremia) and **renal vasoconstriction** (hepatorenal syndrome).
- **Compensated** cirrhosis → **decompensated** when any of: **ascites**, **variceal bleeding**, **hepatic encephalopathy**, jaundice (survival falls sharply).

Severity scores

Score	Components	Interpretation
Child-Pugh	**Bilirubin**, **albumin**, **INR**, **ascites**, **encephalopathy** (mnemonic: **BEAAP** — bilirubin, encephalopathy, ascites, albumin, prothrombin time)	**Class A: 5–6** (compensated) • **B: 7–9** • **C: 10–15** (worst) • Used for **drug dosing** in hepatic impairment
MELD / MELD-Na (MELD 3.0)	**Bilirubin**, **INR**, **creatinine**, **sodium** (± albumin, sex)	Predicts **3-month mortality** • Used to prioritize **liver transplant** • ≥ 15 → refer for transplant

General care in all cirrhosis patients

- **Stop alcohol** completely • treat the cause (HBV, HCV, MASLD/obesity).
- **Vaccinate** against HAV, HBV, pneumococcus, influenza.
- **HCC surveillance:** ultrasound ± AFP every 6 months.
- **Nutrition: protein 1.2–1.5 g/kg/day** (**do not restrict protein**), 35 kcal/kg/day, small frequent meals, **late-evening snack**.
- **Avoid: NSAIDs**, **alcohol**, sedatives/**benzodiazepines**, **aminoglycosides** • acetaminophen up to **2 g/day** is the preferred analgesic.
- Refer to a transplant center when decompensated or MELD ≥ 15.

2. Ascites

Overview and diagnosis

- **Most common** complication of cirrhosis (and the most common first decompensation).
- **Diagnostic paracentesis** for: **new-onset ascites**, **every hospital admission**, and **any clinical deterioration** (fever, pain, encephalopathy, AKI) → to rule out **SBP**.
- Ascitic fluid tests: **cell count with differential**, **albumin**, **total protein**, **culture** (inoculate blood culture bottles at the bedside).
- **SAAG** (serum-ascites albumin gradient) = serum albumin − ascitic albumin • **≥ 1.1 g/dL** → **portal hypertension** (cirrhosis, HF) • **< 1.1** → other causes (malignancy, TB, pancreatitis, nephrotic syndrome).
- **Ascitic total protein < 1.5 g/dL** → higher **SBP** risk.

Grade	Description	Treatment
Grade 1 (mild)	Detected only by ultrasound	Sodium restriction
Grade 2 (moderate)	Moderate abdominal distension	**Sodium restriction + diuretics**

Grade	Description	Treatment
Grade 3 (large/tense)	Marked distension	**Large-volume paracentesis (LVP) + albumin**, then sodium restriction + diuretics

Treatment

Step	Details
1. Sodium restriction	**2 g/day (88 mmol/day)** • Stricter restriction is not recommended (poor nutrition)
2. Fluid restriction	**Only if serum Na < 125 mEq/L** (1–1.5 L/day)
3. Alcohol abstinence	Can greatly improve ascites
4. Diuretics	**Spironolactone 100 mg + furosemide 40 mg** once daily in the **morning** (ratio **100:40** keeps potassium balanced) • Increase together every **3–5 days** → max **spironolactone 400 mg + furosemide 160 mg** • Spironolactone alone may be used for a first, mild episode
Target weight loss	**0.5 kg/day** (no peripheral edema) • **1 kg/day** (with peripheral edema) — faster loss → AKI, hyponatremia
Gynecomastia (spironolactone)	Switch to **amiloride** (10–40 mg/day) or **eplerenone**
Monitoring	**Weight** daily, **Na, K, SCr** • **Urine Na** (or urine Na/K ratio): Na/K > 1 with weight gain → patient is **not following the sodium restriction**

Hold or reduce diuretics if

- Severe **hyponatremia** (Na < 125, especially < 120) • **AKI** (rising SCr) • **hepatic encephalopathy** • **hyperkalemia** (> 6, spironolactone) or **hypokalemia** (< 3, furosemide) • incapacitating **muscle cramps**.

Large-volume paracentesis (LVP)

- **First-line for tense (grade 3) ascites** and refractory ascites.
- If **> 5 L** removed → give **albumin 6–8 g per liter removed** (as 25% albumin) to prevent **post-paracentesis circulatory dysfunction** (hypotension, AKI, hyponatremia, HRS).
- Continue sodium restriction and diuretics after LVP.

Refractory ascites

- Ascites that does not respond to max diuretics + sodium restriction, or recurs quickly, or diuretic complications prevent effective doses.
- **Options: serial LVP + albumin** • **TIPS** (transjugular intrahepatic portosystemic shunt) — risk of **hepatic encephalopathy**; avoid in heart failure, severe pulmonary hypertension, uncontrolled HE • **liver transplant** referral • **midodrine** 7.5 mg TID may help.
- **Beta-blockers** in refractory ascites: **reduce or stop** if **SBP < 90 mmHg**, **Na < 130**, or **AKI**.

Hyponatremia in cirrhosis

- **Hypervolemic** (dilutional) • Treat if Na < 125: **fluid restriction** (1–1.5 L/day), stop diuretics, albumin may help.
- **Tolvaptan** (vasopressin antagonist): **boxed warning — avoid in liver disease** (hepatotoxicity); if used, short-term (≤ 30 days), start in hospital • **Hypertonic saline** only for severe symptoms or before transplant.
- **Correct slowly**: ≤ 8 mEq/L per 24 h → avoid **osmotic demyelination syndrome** (patients with cirrhosis are high risk).

Drugs to AVOID in ascites

- **NSAIDs** → sodium retention, AKI, HRS, diuretic resistance.
- **ACE inhibitors and ARBs** → hypotension, AKI.
- **Alpha-1 blockers** (prazosin) → hypotension.
- **Aminoglycosides** → nephrotoxicity (HRS).
- **PPIs** without clear indication → ↑ SBP and HE risk.

3. Spontaneous Bacterial Peritonitis (SBP)

Overview

- Infection of ascitic fluid **without a surgical source** (bacteria translocate from the gut).
- Organisms: **gram-negative enteric bacteria** (**E. coli**, **Klebsiella**) most common, then streptococci, enterococci.
- **Symptoms:** fever, **abdominal pain/tenderness**, **altered mental status (HE)**, worsening **renal function**, diarrhea, ileus — may be **asymptomatic** → always tap hospitalized patients with ascites.
- **Diagnosis:** ascitic fluid **neutrophil (PMN) count ≥ 250 cells/mm³** (culture may be negative). Start antibiotics immediately — **do not wait** for culture.

Treatment

Component	Details
Empiric antibiotic (community-acquired)	**Cefotaxime 2 g IV q8h** or **ceftriaxone 1–2 g IV daily** for **5 days** • Uncomplicated (no shock, HE, or AKI, no prior quinolone use): **oral ofloxacin/ciprofloxacin** is an option
Nosocomial or recent beta-lactam exposure / high resistance	Broader coverage: **piperacillin-tazobactam** or **carbapenem** ± vancomycin/daptomycin (based on local resistance)
IV albumin (high-yield)	**1.5 g/kg on day 1** (within 6 hours) and **1 g/kg on day 3** — if **SCr > 1 mg/dL**, **BUN > 30 mg/dL**, or **bilirubin > 4 mg/dL** • **↓ hepatorenal syndrome** and **mortality**
Repeat paracentesis	At **48 hours** if no clinical improvement → PMN should fall by ≥ 25%
Other	Hold **NSAIDs**, nephrotoxins • consider stopping **beta-blockers** if hypotensive • avoid LVP during active SBP if possible

SBP prophylaxis

Indication	Regimen
Cirrhosis with GI bleeding (with or without ascites)	**Ceftriaxone 1 g IV daily** for up to **7 days** (switch to oral cipro when able) • **↓** infections, rebleeding, and death
Secondary prophylaxis (after one episode of SBP)	**Long-term (indefinite)**: **ciprofloxacin 500 mg daily** or **TMP-SMX DS 1 tablet daily** (norfloxacin 400 mg daily where available)
Primary prophylaxis (never had SBP)	If **ascitic protein < 1.5 g/dL plus** renal dysfunction (**SCr ≥ 1.2**, BUN ≥ 25, or **Na ≤ 130**) or liver failure (**Child-Pugh ≥ 9** and **bilirubin ≥ 3**) → ciprofloxacin or TMP-SMX daily

- **Daily** dosing (not weekly) → less resistance • Risks: **resistant organisms**, **C. difficile** • Stop when ascites resolves or after transplant.

4. Gastroesophageal Varices and Variceal Bleeding

Overview

- Dilated collateral veins (esophagus, stomach) due to **portal hypertension** (HVPG ≥ 10) • bleed when HVPG ≥ 12.
- **Variceal hemorrhage** → high mortality (~15–20% at 6 weeks).
- **Screening:** upper endoscopy (EGD) at diagnosis of cirrhosis • may be avoided if **liver stiffness < 20 kPa** and **platelets > 150,000** (Baveno VII).
- **Red wale signs** (red marks on varices), large size, and **Child-Pugh C** → high bleeding risk.

Non-selective beta-blockers (NSBBs)

- **Mechanism: β1 blockade** → ↓ cardiac output • **β2 blockade** → unopposed α-constriction of splanchnic vessels → **↓ portal blood flow and pressure**. **Carvedilol** also blocks **α1** → ↓ intrahepatic resistance (**greater portal pressure reduction**).

- **Selective β1-blockers** (metoprolol, atenolol) are **NOT effective**.
- **Titrate** to **resting HR 55–60 bpm** while keeping **SBP ≥ 90 mmHg** • continue **indefinitely**.

Drug	Dose	Notes
Carvedilol (preferred in many guidelines)	**6.25 mg once daily** → after 3 days **6.25 mg BID** • max **12.5 mg/day**	Best portal pressure reduction • also prevents decompensation in clinically significant portal hypertension • more hypotension
Propranolol	**20–40 mg BID**, titrate every 2–3 days • max **160 mg/day** (**80 mg/day** with ascites)	Hepatic metabolism
Nadolol	**20–40 mg once daily** • max **160 mg/day** (**80 mg/day** with ascites)	**Renal elimination** (adjust in CKD)

- **ADRs:** fatigue, **hypotension**, **bradycardia**, dizziness, bronchospasm, erectile dysfunction, may **mask hypoglycemia** • **Contraindications:** asthma, heart block, severe bradycardia, decompensated HF • **Do not stop abruptly**.
- **Reduce or stop** NSBB if: **SBP < 90**, **Na < 130**, **AKI**, **SBP infection with hypotension**, or active bleeding with shock.
- **Nitrates** (isosorbide mononitrate) are **not recommended** (alone or added).

Primary prophylaxis (never bled)

Situation	Prophylaxis
Medium/large varices	**NSBB** (carvedilol, propranolol, nadolol) **OR endoscopic variceal ligation (EVL)** — equally effective; **do not combine** for primary prophylaxis
Small varices with red wale signs or Child-Pugh C	**NSBB**
Small varices, low risk	NSBB optional • repeat EGD
Clinically significant portal hypertension (no varices needing treatment)	**NSBB (carvedilol)** to prevent decompensation (Baveno VII)

Acute variceal hemorrhage (AVH) — management steps

1. **Resuscitation:** airway protection (intubation if needed), IV access • **Restrictive transfusion:** transfuse at **Hgb < 7 g/dL**, target **7–9 g/dL** (**over-transfusion ↑ portal pressure** → rebleeding) • avoid excessive saline • FFP/platelets not routinely needed.
2. **Vasoactive drug as early as possible** (before endoscopy), continue **2–5 days**: **octreotide 50 mcg IV bolus**, then **50 mcg/h infusion** (US) • **terlipressin** (outside US: 2 mg IV q4h × 48 h, then 1 mg q4h) • vasopressin + nitroglycerin (rarely).
3. **Antibiotic prophylaxis: ceftriaxone 1 g IV daily** (max **7 days**).
4. **Upper endoscopy within 12 hours** → **endoscopic variceal ligation (EVL / banding)** for esophageal varices • gastric varices: **cyanoacrylate glue**, BRTO, or TIPS.
5. **Pre-emptive TIPS** (within 72 h, ideally < 24 h) for high-risk patients: **Child-Pugh C (10–13)** or **Child-Pugh B > 7 with active bleeding** at endoscopy.
6. **Rescue** for uncontrolled bleeding: **balloon tamponade** (bridge, ≤ 24 h) or esophageal stent → **rescue TIPS**.
7. **Hold NSBBs** during the acute bleed (hypotension) • **lactulose** to prevent HE (blood is a protein load in the gut) • short-course **PPI** after EVL (↓ post-banding ulcers).
8. **Avoid** NSAIDs, sedatives; correct coagulopathy only when clinically indicated (INR does not predict bleeding in cirrhosis).

Secondary prophylaxis (after a bleed)

- **Combination: NSBB** (propranolol or nadolol; carvedilol alternative) **+ EVL** repeated every **1–4 weeks** until varices are obliterated, then surveillance EGD.
- Start NSBB once the patient is stable and vasoactive drug is stopped (usually **day 3–6**).
- **TIPS** if rebleeding despite NSBB + EVL • liver transplant evaluation.

5. Hepatic Encephalopathy (HE)

Overview

- Brain dysfunction caused by **liver insufficiency and/or portosystemic shunting** → toxins (mainly **ammonia** from gut bacteria breaking down protein) reach the brain; also inflammation, GABA-like substances, manganese.
- **Asterixis** (flapping tremor) is a classic sign.
- **Ammonia level**: **not needed** for diagnosis or monitoring (does not correlate well with severity); a **normal** level makes HE unlikely.
- HE is a **clinical diagnosis** → rule out other causes (hypoglycemia, infection, stroke, drugs, alcohol withdrawal, Wernicke's).

West Haven grading

Grade	Features	Type
Minimal	Normal exam; abnormal only on psychometric/neuropsychological tests (driving risk)	**Covert**
Grade 1	Mild confusion, ↓ attention, anxiety/euphoria, **sleep-wake reversal**, impaired calculation	**Covert**
Grade 2	**Lethargy**, **disorientation to time**, personality change, inappropriate behavior, **asterixis**	**Overt**
Grade 3	**Somnolence** to semi-stupor (responds to stimuli), gross disorientation, confusion	**Overt**
Grade 4	**Coma**	**Overt**

Precipitating factors (identify and correct FIRST)

Category	Examples
Infection	**SBP**, UTI, pneumonia, sepsis (most common)
GI bleeding	Variceal or ulcer bleeding (blood = protein load)
Constipation	↑ ammonia absorption • **non-adherence to lactulose**
Dehydration / electrolytes	**Over-diuresis**, vomiting, diarrhea • **hypokalemia** (↑ renal ammonia production) • hyponatremia • metabolic **alkalosis** (converts $NH_4^+ \rightarrow NH_3$, which crosses into the brain)
Drugs	**Benzodiazepines**, **opioids**, sedatives, sedating antihistamines, anticholinergics, **alcohol**
Other	**AKI**, hypoglycemia, **TIPS**, surgery, hypoxia, excessive dietary protein (rare)

Treatment of overt HE

Drug	Details
Lactulose — first-line	**Mechanism:** colonic bacteria break lactulose into acids → **acidify the colon** → **NH3 converted to NH4+** (trapped and excreted) + **osmotic laxative** (↓ transit time) + changes gut flora • **Dose: 25 mL (≈ 16.7 g) every 1–2 hours** until ≥ 2 soft bowel movements, then **titrate to 2–3 soft stools/day** (usually 15–30 mL BID–TID) • **Retention enema** (300 mL in 700 mL water) if unable to take oral • **ADRs: diarrhea** (→ dehydration, **hypernatremia**, **hypokalemia** → can **worsen HE**), bloating, flatulence, cramps, very sweet taste • **Contraindicated:** galactosemia
Rifaximin — add-on	**550 mg BID** • **Minimally absorbed** antibiotic → ↓ ammonia-producing gut bacteria • **Add to lactulose after a recurrence** (prevents HE episodes and hospitalizations) • Can be used if lactulose not tolerated • ADRs: nausea, peripheral edema, dizziness, **C. difficile** (low) • Expensive • P-gp substrate (cyclosporine ↑ absorption)
Polyethylene glycol 3350	**4 L over 4 hours** — alternative to lactulose in acute HE (faster resolution in some trials)
Neomycin	1–4 g/day in divided doses • rarely used: **ototoxicity**, **nephrotoxicity** (some absorption)
Metronidazole	250 mg BID, **short-term only** • **peripheral neuropathy**, neurotoxicity
Others	**Zinc** (if deficient) • **L-ornithine L-aspartate (LOLA)** • **oral branched-chain amino acids (BCAA)** for protein-intolerant patients • **Flumazenil** only if benzodiazepine-related (temporary)

Prevention

- **After the first episode of overt HE → lactulose** (secondary prophylaxis).
- **After a recurrence → lactulose + rifaximin**.
- **Primary prophylaxis** not routinely recommended (except during variceal bleeding, lactulose may be used).
- **Minimal/covert HE**: lactulose or rifaximin may be considered; counsel about **driving**.
- Continue therapy indefinitely unless precipitant fully controlled or after liver transplant.

Nutrition and drug safety in HE

- **Do NOT restrict protein → 1.2–1.5 g/kg/day** (protein restriction worsens malnutrition and muscle loss; muscle helps clear ammonia).
- Small frequent meals, **late-night snack**; **vegetable and dairy protein** better tolerated than meat.
- **Avoid benzodiazepines and opioids**; if a benzodiazepine is unavoidable, use one without active metabolites (**lorazepam, oxazepam**) at low dose.
- **Liver transplant** is the definitive treatment for recurrent HE.

6. Hepatorenal Syndrome (HRS-AKI)

Overview

- **Functional** kidney failure in advanced cirrhosis with ascites: severe **renal vasoconstriction** (from splanchnic vasodilation) — the kidneys are **structurally normal** (would work if transplanted).
- **Diagnosis of exclusion** • looks like **prerenal AKI**: **low urine Na (< 10)**, **FENa < 1%**, bland sediment, but **no response to volume**.
- **Types: HRS-AKI** (formerly type 1: rapid, often triggered by SBP) • **HRS-NAKI/CKD** (formerly type 2: slower, with refractory ascites).
- **Precipitants: SBP** (most common) and other infections, **GI bleeding**, **LVP without albumin**, over-diuresis, **NSAIDs**.

Diagnostic criteria (ICA-ADQI)

1. **Cirrhosis with ascites**.
2. **AKI**: SCr ↑ **≥ 0.3 mg/dL within 48 h** or **≥ 50% within 7 days**.

3. **No response** after **2 consecutive days** of **diuretic withdrawal** + volume expansion with **albumin 1 g/kg/day** (max 100 g/day).
4. **No shock**.
5. **No current or recent nephrotoxic drugs** (NSAIDs, aminoglycosides, iodinated contrast).
6. **No signs of structural kidney injury**: proteinuria < 500 mg/day, no microhematuria (< 50 RBC/hpf), normal kidney ultrasound.

Management

Step	Details
1. Stop offending drugs	**Diuretics**, **NSAIDs**, **ACEi/ARBs**, nephrotoxins • consider holding **NSBBs** (if hypotensive) • treat infection
2. Albumin challenge	**1 g/kg/day IV (max 100 g/day) × 2 days**
3. Vasoconstrictor + albumin (first-line)	**Terlipressin + albumin** (20–40 g/day) • ICU: **norepinephrine + albumin** • If neither available: **midodrine + octreotide + albumin** (least effective)
4. Bridge / definitive	**Dialysis** (bridge to transplant) • TIPS in selected patients • **Liver transplant** = **definitive** treatment (± simultaneous liver-kidney)

Vasoconstrictors

Drug	Dose	Key points
Terlipressin (Terlivaz) — vasopressin analog (V1)	**1 mg IV q4–6h** (label: 0.85 mg q6h) • ↑ to **2 mg q4–6h** if SCr ↓ < 25% by day 4 • up to **14 days**	**Boxed warning: serious or fatal respiratory failure** → avoid if **SpO2 < 90%** or **volume overload** (manage fluids, reduce albumin), monitor oxygen • Less effective if **SCr ≥ 5 mg/dL** • ADRs: **abdominal pain**, nausea, **diarrhea**, **ischemia** (cardiac, mesenteric, peripheral/digital), bradycardia, hyponatremia
Norepinephrine	**0.5–3 mg/h IV** continuous (ICU) → goal: ↑ **MAP by 10 mmHg**	Requires central line and ICU • arrhythmias, ischemia
Midodrine + octreotide	Midodrine **7.5 mg PO TID** (↑ to 12.5–15 mg TID) + octreotide **100 mcg SC TID** (↑ to 200 mcg TID)	Outside ICU when terlipressin/NE unavailable • midodrine: supine hypertension, urinary retention, piloerection

Prevention of HRS

- **Albumin** with **SBP** treatment (1.5 g/kg day 1, 1 g/kg day 3) • **albumin with LVP > 5 L**.
- **Avoid NSAIDs**, aminoglycosides, contrast, over-diuresis • SBP prophylaxis in high-risk patients.

7. Other Complications (brief)

Complication	Key points
Coagulopathy	↑ INR does **not** reflect bleeding risk (rebalanced hemostasis; cirrhosis patients can still clot → **VTE prophylaxis** when hospitalized) • **Vitamin K** only if deficiency suspected (cholestasis, malnutrition)
Thrombocytopenia	From splenic sequestration and ↓ thrombopoietin • before procedures: **avatrombopag** or **lusutrombopag** (TPO agonists)
Hepatocellular carcinoma	**Ultrasound ± AFP every 6 months**
Hepatopulmonary syndrome	Hypoxemia from intrapulmonary vasodilation (platypnea) → oxygen, transplant
Portopulmonary hypertension	Pulmonary arterial hypertension + portal hypertension • avoid beta-blockers
Hepatic hydrothorax	Usually right-sided pleural effusion • sodium restriction + diuretics, thoracentesis, TIPS • **no chest tube**
Pruritus (cholestatic)	**Cholestyramine** (separate other drugs), rifampin, naltrexone, sertraline
Muscle cramps	Correct electrolytes, reduce diuretics, albumin, **baclofen**, taurine
Malnutrition/sarcopenia	Protein 1.2–1.5 g/kg/day, late-evening snack, exercise

8. Medication Use in Cirrhosis (high-yield)

Drug / class	Recommendation
Acetaminophen	**Preferred analgesic** • up to **2 g/day**
NSAIDs	**Avoid** (AKI, HRS, GI bleeding, diuretic resistance)
Opioids	Avoid if possible (HE, constipation) • if needed: low-dose **hydromorphone** or **fentanyl** • **avoid codeine, meperidine, tramadol** (unpredictable/active metabolites, seizures)
Benzodiazepines / sedatives	**Avoid** (precipitate HE) • if necessary: lorazepam, oxazepam (low dose)
ACEi / ARBs	**Avoid in ascites** (hypotension, AKI)
Aminoglycosides	**Avoid** (nephrotoxicity, HRS)
PPIs	Use only with a clear indication (↑ **SBP**, **HE**, C. diff)
Metformin	Avoid in **decompensated** cirrhosis (lactic acidosis)
Statins	Safe in **compensated** cirrhosis (may even be beneficial) • avoid/caution in decompensated (Child-Pugh C)
Dose adjustment	Use **Child-Pugh class** for drugs with hepatic clearance; reduce doses of high-extraction drugs

Exam Traps and Frequently Tested Points

1. **Child-Pugh:** bilirubin, albumin, INR, ascites, encephalopathy (A 5–6, B 7–9, C 10–15) • **MELD:** bilirubin, INR, creatinine, Na → transplant priority.
2. **Ascites:** sodium **2 g/day**; **fluid restriction only if Na < 125**.
3. **Diuretics: spironolactone 100 mg + furosemide 40 mg** (100:40 ratio), max **400:160**, titrate every 3–5 days.
4. **Weight loss goal:** 0.5 kg/day (no edema), 1 kg/day (edema).
5. **Gynecomastia** from spironolactone → **amiloride** or eplerenone.
6. **Tense ascites → LVP • > 5 L removed → albumin 6–8 g/L**.
7. **SAAG ≥ 1.1** = portal hypertension.
8. **Avoid in ascites:** NSAIDs, ACEi/ARBs, aminoglycosides • stop NSBB if SBP < 90, Na < 130, or AKI.
9. **Tolvaptan** → boxed warning: avoid in liver disease • correct Na ≤ 8 mEq/L/24 h.
10. **SBP diagnosis: ascitic PMN ≥ 250 cells/mm³** • most common organism **E. coli**.
11. **SBP treatment: cefotaxime 2 g q8h** or **ceftriaxone** × 5 days + **albumin 1.5 g/kg day 1 and 1 g/kg day 3** (if SCr > 1, BUN > 30, or bilirubin > 4).
12. **Albumin in SBP ↓ HRS and mortality.**
13. **GI bleed in cirrhosis → ceftriaxone 1 g daily ≤ 7 days.**
14. **After SBP → lifelong ciprofloxacin or TMP-SMX daily** (secondary prophylaxis).
15. **Primary SBP prophylaxis:** ascitic protein < 1.5 + renal dysfunction or liver failure.
16. **NSBBs** (propranolol, nadolol, carvedilol) → HR **55–60**, SBP ≥ 90 **• selective β1-blockers don't work • no nitrates**.
17. **Carvedilol** → greatest portal pressure ↓ (α1 + β), max 12.5 mg/day.
18. **Primary prophylaxis (medium/large varices):** NSBB **or** EVL (not both).
19. **Secondary prophylaxis:** NSBB **+** EVL (both).
20. **Acute variceal bleed: octreotide** (50 mcg bolus + 50 mcg/h × 2–5 days) + **ceftriaxone** + **EVL within 12 h** + **restrictive transfusion (Hgb 7) • hold NSBB** during the bleed.
21. **Pre-emptive TIPS** within 72 h for Child C 10–13 or Child B with active bleeding.
22. **TIPS** ↑ hepatic encephalopathy risk.
23. **HE: find the precipitant first** (infection, GI bleed, constipation, dehydration, hypokalemia, benzodiazepines/opioids).
24. **Ammonia levels not needed** for diagnosis or monitoring.

25. **Lactulose** first-line → titrate to **2–3 soft stools/day** • too much → dehydration, hypernatremia → worse HE.
26. **Rifaximin 550 mg BID** → add to lactulose for **recurrent** HE.
27. **Do NOT restrict protein in HE** (1.2–1.5 g/kg/day).
28. **Hypokalemia and alkalosis worsen HE** (↑ ammonia).
29. **HRS:** functional, kidneys normal, **no response to albumin 1 g/kg × 2 days**, urine Na < 10, no nephrotoxins, no shock.
30. **HRS treatment: terlipressin + albumin** (boxed: **respiratory failure**; avoid if SpO2 < 90% or volume overload) • norepinephrine + albumin (ICU) • midodrine + octreotide + albumin • **liver transplant is definitive**.
31. **Most common HRS precipitant: SBP.**
32. **Acetaminophen ≤ 2 g/day** is the safest analgesic; **avoid NSAIDs**.
33. **INR does not predict bleeding** in cirrhosis; give VTE prophylaxis when hospitalized.

Note: Doses and recommendations differ slightly between guidelines (AASLD, EASL, Baveno VII). **Use the numbers in the reference you are studying for the exam.**

Drug Summary Table: Doses, Key ADRs and DDIs

Drug	Dose	Key ADRs	Key DDIs / special notes
Spironolactone	100 mg daily (AM) → max 400 mg/day (with furosemide 100:40)	**Hyperkalemia**, **gynecomastia**, AKI, hyponatremia	↑ K with ACEi/ARBs, K supplements, TMP-SMX, salt substitutes • slow onset (days)
Furosemide	40 mg daily (AM) → max 160 mg/day	**Hypokalemia**, hyponatremia, AKI, ototoxicity, cramps	Hypokalemia worsens **HE** • ↑ ototoxicity with aminoglycosides • NSAIDs ↓ effect
Amiloride	10–40 mg/day	Hyperkalemia	Alternative if gynecomastia (less effective)
Eplerenone	25–50 mg daily	Hyperkalemia	No gynecomastia • **CYP3A4 substrate** (avoid strong inhibitors)
Albumin 25%	LVP: 6–8 g/L removed (> 5 L) • SBP: 1.5 g/kg day 1, 1 g/kg day 3 • HRS: 1 g/kg/day × 2 (max 100 g), then 20–40 g/day	**Volume overload**, pulmonary edema	Watch fluid status, especially with **terlipressin**
Tolvaptan	15 mg daily (≤ 30 days, start in hospital)	**Boxed: hepatotoxicity (avoid in liver disease), osmotic demyelination** (rapid Na rise)	**CYP3A4 substrate** (CI with strong inhibitors) • avoid fluid restriction in first 24 h
Midodrine	7.5 mg TID (up to 12.5–15 mg TID)	**Supine hypertension**, urinary retention, piloerection, bradycardia	↑ BP with other vasoconstrictors • last dose before 6 pm
Cefotaxime	2 g IV q8h × 5 days (SBP)	Diarrhea, rash, C. diff	First-line SBP treatment
Ceftriaxone	SBP: 1–2 g IV daily × 5 days • GI bleed prophylaxis: 1 g daily ≤ 7 days	Diarrhea, biliary sludge, C. diff	**Calcium-containing IV solutions** in neonates (CI)
Ciprofloxacin	500 mg daily (SBP prophylaxis)	Tendinopathy, **QT**, C. diff, dysglycemia	**Chelation** (antacids, iron, Ca) • ↑ theophylline, **tizanidine (CI)**, warfarin
TMP-SMX DS	1 tablet daily (SBP prophylaxis)	Rash/SJS, **hyperkalemia**, cytopenias, ↑ SCr	↑ K with spironolactone • ↑ warfarin INR, ↑ methotrexate toxicity
Norfloxacin	400 mg daily (where available)	Similar to fluoroquinolones	Not available in the US

Drug	Dose	Key ADRs	Key DDIs / special notes
Carvedilol	6.25 mg daily → 6.25 mg BID (max 12.5 mg/day)	**Hypotension**, bradycardia, fatigue, dizziness	Stop/reduce if SBP < 90, Na < 130, AKI • masks hypoglycemia • additive with other antihypertensives
Propranolol	20–40 mg BID, titrate (max 160 mg/day; 80 mg with ascites)	Bradycardia, hypotension, fatigue, bronchospasm, ED	Target HR 55–60 • avoid in asthma • don't stop abruptly
Nadolol	20–40 mg daily (max 160 mg/day; 80 mg with ascites)	Same as propranolol	**Renally cleared** → adjust in CKD
Octreotide	Variceal bleed: 50 mcg IV bolus + 50 mcg/h × 2–5 days • HRS: 100–200 mcg SC TID	**Hyper/hypoglycemia**, bradycardia, GI upset, gallstones (long term)	Start before endoscopy • ↓ cyclosporine absorption
Terlipressin	1 mg IV q4–6h (↑ to 2 mg) up to 14 days (HRS) • 2 mg q4h (variceal bleed, outside US)	**Boxed: fatal respiratory failure • ischemia**, abdominal pain, diarrhea, bradycardia, hyponatremia	Avoid if **SpO2 < 90%**, volume overload • less effective if SCr ≥ 5
Vasopressin	0.2–0.4 units/min IV (with nitroglycerin)	**Ischemia** (cardiac, mesenteric), hyponatremia	Rarely used now
Norepinephrine	0.5–3 mg/h IV (HRS, ICU)	Arrhythmias, ischemia, extravasation necrosis	Central line • ↑ effect with MAOIs, TCAs
Lactulose	Acute: 25 mL q1–2h until 2 BMs, then 15–30 mL BID–TID (goal 2–3 soft stools/day) • enema 300 mL in 700 mL water	**Diarrhea**, dehydration, **hypernatremia**, **hypokalemia**, bloating	CI in **galactosemia** • antacids may ↓ colonic acidification
Rifaximin	550 mg BID	Nausea, **peripheral edema**, dizziness, C. diff (low)	Minimal absorption • **P-gp inhibitors** (cyclosporine) ↑ levels • use with lactulose
PEG 3350	4 L over 4 h (acute HE alternative)	Bloating, nausea, electrolyte shifts	Alternative to lactulose
Neomycin	1–4 g/day divided (short term)	**Ototoxicity**, **nephrotoxicity**, malabsorption	Avoid in renal impairment • rarely used
Metronidazole	250 mg BID (short term, HE)	**Peripheral neuropathy**, metallic taste	No alcohol • ↑ warfarin INR
Zinc sulfate	220 mg BID (if deficient)	GI upset, copper deficiency (long term)	↓ absorption of quinolones, tetracyclines
Cholestyramine	4 g 1–4 times daily (pruritus)	Constipation, bloating	Binds other drugs → give 1 h before / 4–6 h after • ↓ fat-soluble vitamins
Avatrombopag / lusutrombopag	Short course before a procedure	Headache, thrombosis (portal vein)	Raise platelets in thrombocytopenia

Good luck on your exam!